

Expanding the control over the operating system from the database

Expanding the control over the operating system from the database - Bernardo Damele A. G., Slideshare.

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![Expanding the control over the operating system from the database Bernardo Damele Assumpção Guimarães Guido Landi

Barcelona (Spain) – September 21, 2009](https://image.slidesharecdn.com/source-barcelona-2009-damele-landi-expanding-the-control-090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-1-320.jpg)

![Who we are

Bernardo Damele Assumpção Guimarães Proud father Penetration tester / security researcher at Portcullis Computer Security Ltd sqlmap lead developer

Guido Landi Reverse engineer Exploit writer Vulnerability researcher SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 2](https://image.slidesharecdn.com/source-barcelona-2009-damele-landi-expanding-the-control-090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-2-320.jpg)

![Introduction

Database management systems are powerful applications

Store and interact with data

Interact with the file system and operating system

When they can't by design, you can force them to When they can't due to limited user's privileges, you can exploit them!

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![Scenario

You have got access to a database Direct access – provided account, weak passwords, brute-forcing credentials SQL injection – web application, stand-alone client, cash machine ☺, ...

What to do now other than enumerating data? Own the underlying operating system Why not even other servers within the DMZ?

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! [State of art – File system access

Microsoft SQL Server Read: BULK INSERT Write: xp_cmdshell / debug.exe

MySQL Read: LOAD_FILE() Write: SELECT ... INTO DUMPFILE

PostgreSQL Read: COPY / UDF Write: Large object's lo_export() SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 5](<https://image.slidesharecdn.com/source-barcelona-2009-damele-landi-expanding-the-control-090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-5-320.jpg>)

! [State of art – Command execution

Microsoft SQL Server OPENROWSET can be abused to escalate privileges Built-in xp_cmdshell to execute commands

Oracle If you find a SQL injection in a function owned by SYS and with authid definer, you can run PL/SQL as SYS Many ways to execute commands. Example: DBMS_EXPORT_EXTENSION package's GET_DOMAIN_INDEX_TABLES() function

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! [State of art – Command execution

MySQL and PostgreSQL support user-defined functions: custom function that can be evaluated in SQL statements

UDF can be created from shared libraries that are compiled binary files Dynamic-link library on Windows Shared object on Linux

PostgreSQL supports also procedural languages SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 7](<https://image.slidesharecdn.com/source-barcelona-2009-damele-landi-expanding-the-control-090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-7-320.jpg>)

! [Demonstration

Operating system command execution by exploiting a SQL injection vulnerability in a web application

Further information on these techniques can be found on <http://tinyurl.com/sqlmap1>

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! [More than command execution

Owning the underlying operating system is not only about command execution

Full-duplex connection between the attacker host and the database server

Database used as a stepping stone to establish this covert channel Shell, Meterpreter, VNC – <http://metasploit.com> DNS tunnel – <http://heyoka.sourceforge.net> SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 9](<https://image.slidesharecdn.com/source-barcelona-2009-damele-landi-expanding-the-control-090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-9-320.jpg>)

![Establish the channel

On your box Forge a stand-alone payload stager with msfpayload Encode it with msfencode to bypass AV Run msfcli with multi/handler exploit

On the database server Upload it to the file system temporary folder Execute it via UDF, xp_cmdshell, ... SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 10] (<https://image.slidesharecdn.com/source-barcelona-2009-damele-landi-expanding-the-control-090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-10-320.jpg>)

![Getting stealth

Anti-forensics technique as an option to upload the stand-alone payload stager executable

On your box Forge a shellcode with msfpayload Encode it with msfencode Run msfcli with multi/handler exploit

On the database Create a UDF that executes a payload in-memory Execute the UDF providing the payload as a parameter SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 11] (<https://image.slidesharecdn.com/source-barcelona-2009-damele-landi-expanding-the-control-090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-11-320.jpg>)

![User-defined function sys_bineval()

Execute an arbitrary payload from the database management system memory

Features

Works in DEP/NX-enabled systems Supports alphanumeric payloads Protects the DBMS if the payload crashes It does not fork a new process

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![sys_bineval() vs DEP/NX Use VirtualAlloc() to allocate an +RWX memory region

```
code = (char *) VirtualAlloc(NULL, 4096, MEM_RESERVE|MEM_COMMIT, PAGE_EXECUTE_READWRITE);
```

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090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-13-320.jpg)

![[sys_bineval() and alphanum payloads Metasploit's msfencode has alphanumeric encoders to encode the payload

Problem: It is not able to produce pure alphanumeric payloads due to get_pc()

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![[sys_bineval() and alphanum payloads Solution: Use the BufferRegister option ./msfencode BufferRegister=EAX -e x86/alpha_mixed ...

Put the payload address in EAX register __asm { MOV EAX, [lpPayload] CALL EAX }

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![[sys_bineval(): avoid DBMS crash Spawn a new thread WaitForSingleObject(CreateThread(NULL, 0, ExecPayload, CodePointer, 0, &pID), INFINITE);

Wrap the payload in a SEH frame __try { __asm { MOV EAX, [lpPayload] CALL EAX } } SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 16](<https://image.slidesharecdn.com/source-barcelona-2009-damele-landi-expanding-the-control-090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-16-320.jpg>)

![[Demonstration

Exploit a SQL injection vulnerability in a web application to establish an out-of-band channel in-memory via a custom UDF

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![[Hands on the Windows registry

Microsoft SQL Server

Built-in stored procedures, xp_reg(read|write|delete)

MySQL and PostgreSQL

Upload and execute a bat file that executes reg (query|add|delete) Upload and execute a file and pass it to regedit SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 18]

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![[MS09-004: Memory corruption

Discovered by Bernhard Mueller, it affects Microsoft SQL Server up to 2005 SP2 Triggered by a call to `sp_replwritetovarbin` No authentication and no privileges needed

"Limited Memory Overwrite Vulnerability" could allow remote code execution "Limited Memory Overwrite"... Actually a pretty huge heap-based buffer overflow, ~4000 bytes

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![Exploiting MS09-004

Target heap metadata Could be hard/unreliable even if Microsoft SQL Server uses a custom allocator

Target application specific data Function pointers, C++ object pointers, etc.

Luckily for us Microsoft SQL Server tries hard to not crash by graceful handling exceptions...

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![Exploiting MS09-004

...this gives us more than one code path to achieve code execution: An almost arbitrary 4-bytes overwrite: `MOV DWORD PTR DS:[EAX+4], EDI`

An object pointer overwrite: `MOV EDX,DWORD PTR DS:[ESI] [...] MOV EAX,DWORD PTR DS:[EDX+10] [...]`

`CALL EAX` SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 21]

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![Bypass hardware-enforced DEP

Use `ret2libc` to call `ZwSetInformationProcess()` Make ESP point to our buffer: `PUSH ESI POP ESP RET`

No need for a fake stack frame, just return in the middle of `LdrpCheckNXCompatibility()`

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![Bypass hardware-enforced DEP `LdrpCheckNXCompatibility()` [...] `MOV DWORD PTR SS:[EBP-4],2 PUSH 4 LEA EAX,DWORD PTR SS:[EBP-4] PUSH EAX PUSH 22 PUSH -1 CALL ntdll.ZwSetInformationProcess [...]`

DEP is now disabled for the current process ...then jump to the shellcode. Game over? SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 23](<https://image.slidesharecdn.com/source-barcelona-2009-damele-landi-expanding-the-control-090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-23-320.jpg>)

![Avoid crash

The original stack address is gone, ESP and EBP point to our buffer

Even if Microsoft SQL Server tries hard to handle exceptions, it will eventually crash

We need to restore ESP and EBP

Is there a generic way?

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![Thread Environment Block

TEB stores information about the currently running thread: 0:000> !teb TEB at 7ffdd000 ExceptionList: 0012fd04 StackBase: 00130000 StackLimit: 0012e000 SubSystemTib: 00000000 FiberData: 00001e00 ArbitraryUserPointer: 00000000 Self: 7ffdd000 EnvironmentPointer: 00000000 ClientId: 00000d2c RpcHandle: 00000000 Tls Storage: 00000000 [...] SOURCE Conference 2009, Barcelona (Spain) September 21, 2009 25](<https://image.slidesharecdn.com/source-barcelona-2009-damele-landi-expanding-the-control-090924105645-phpapp01/85/Expanding-the-control-over-the-operating-system-from-the-database-25-320.jpg>)

![TEB: Restore the Stack Pointer(s)

Contains 3 pointers to the current thread's stack

Addressable through the FS segment register

Just prepend the shellcode with a little stub: MOV ESP, DWORD PTR FS:[0] MOV EBP, ESP SUB ESP, 20
Game Over!

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![Demonstration

Own the system by exploiting MS09-004 vulnerability via a SQL injection vulnerability in a web application with back-end Microsoft SQL Server

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![But... Wasn't it meant to deal with data? Once you get access to a database you can compromise the whole system in most cases With a comfortable, fast and stable channel

Once you have access to the system you can escalate privileges (token kidnapping, software bugs, kernel flaws, weak privileges, etc.)

When you are root/Administrator/SYSTEM you can crack users' passwords or impersonate them to get access to other servers within the network perimeter

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! [Credits

Our mums for buying our first pre-school computers

Alessandro Tanasi and Oliver Gruskovnjak for the technical discussions

skape and skywing for their paper on DEP bypass

H D Moore and the Metasploit development team

Stacy Thayer and the SOURCE Conference team

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! [Questions?

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! [Thanks for your attention!

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